MILITARY SPECIFICATION SHEET

STUD, SELF-LOCKING, FLUSH AND PROTRUDING HEAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the studs described herein shall consist of this specification and the latest issue of MIL-S-63540.

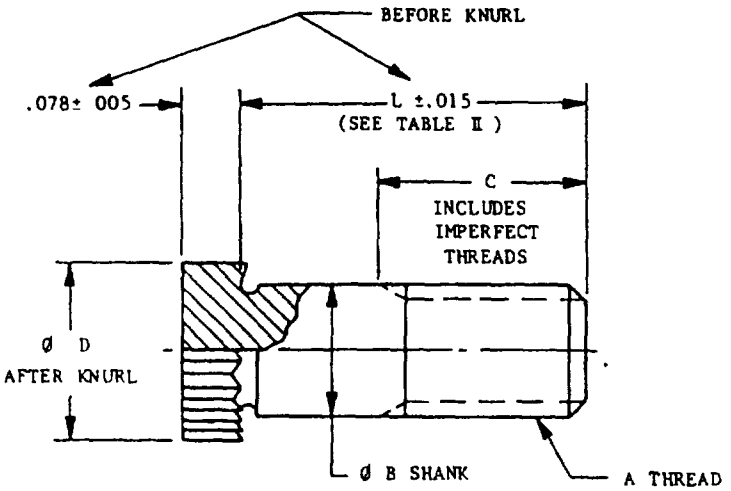


FIGURE 1. Stud, long length

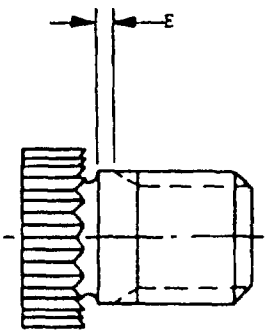


FIGURE 2 Stud, short length
NOTE Details same as figure 1 except for dimension "E". "C" thread length is not applicable

TABLE I: Part dimensions

A THREAD CLASS-3A	Ø B +.000 -.005	C ±.015	Ø D ±.015	E +.015 -.010
.086-64UNF	.086	.500	.170	.023
.086-56UNC				
.112-48UNF	.112			.030
.112-40UNC				
.138-40UNF	.138		.200	.036
.138-32UNC				
.164-36UNF	.164		.220	
.164-32UNC				
.190-32UNF	.190	.620	.260	.047
.190-24UNC				
.250-28UNF	.250		.320	.055
.250-20UNC				

NOTES:
1. All dimensions are in inches.

FSC 5307

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TABLE II: Thread size and length dash numbers

L	THREAD SIZE											
	.086-64 UNF	.086-56 UNC	112-48 UNF	.112-40 UNC	138-40 UNF	138-32 UNC	164-36 UNF	.164-32 UNC	190-32 UNF	.190-24 UNC	250-28 UNF	250-20 UNC
±.015	THREAD SIZE AND LENGTH DASH NUMBER											
.250	01E4	02E4										
312	01E5	02E5	03E5	04E5								
.375	01E6	02E6	03E6	04E6	05E6	06F6						
.438	01E7	02E7	03E7	04E7	05E7	06E7	07E7	08E7	09E7	10E7		
.500	01E8	02E8	03E8	04E8	05F8	06F8	07E8	08E8	09E8	10E8		
562	01E9	02E9	03E9	04E9	05E9	06F9	07F9	08F9	09E9	10E9		
.625	01-10	02-10	03-10	04-10	05-10	06-10	07-10	08-10	09E10	10E10	11E10	12E10
.688			03-11	04-11	05-11	06-11	07-11	08-11	09E11	10E11	11E11	12E11
750			03-12	04-12	05-12	06-12	07-12	08-12	09-12	10-12	11-12	12-12
.812					05-13	06-13	07-13	08-13	09-13	10-13	11-13	12-13
.875					05-14	06-14	07-14	08-14	09-14	10-14	11-14	12-14
.938					05-15	06-15	07-15	08-15	09-15	10-15	11-15	12-15
1.000					05-16	06-16	07-16	08-16	09-16	10-16	11-16	12-16
1.062					05-17	06-17	07-17	08-17	09-17	10-17	11-17	12-17
1.125					05-18	06-18	07-18	08-18	09-18	10-18	11-18	12-18
1.188					05-19	06-19	07-19	08-19	09-19	10-19	11-19	12-19
1.250					05-20	06-20	07-20	08-20	09-20	10-20	11-20	12-20
1.312							07-21	08-21	09-21	10-21	11-21	12-21
1.375							07-22	08-22	09-22	10-22	11-22	12-22
1.438							07-23	08-23	09-23	10-23	11-23	12-23
1.500							07-24	08-24	09-24	10-24	11-24	12-24
1.562									09-25	10-25	11-25	12-25
1.625									09-26	10-26	11-26	12-26
1.688									09-27	10-27	11-27	12-27
1.750									09-28	10-28	11-28	12-28
1.812									09-29	10-29	11-29	12-29
1.875									09-30	10-30	11-30	12-30
1.938									09-31	10-31	11-31	12-31
2 000									09-32	10-32	11-32	12-32
2 062											11-33	12-33
2.125											11-34	12-34
2.188											11-35	12-35
2.250											11-36	12-36
2.312											11-37	12-37
2.375											11-38	12-38
2.438											11-39	12-39
2.500											11-40	12-40
2.562											11-41	12-41
2.625											11-42	12-42
2.688											11-43	12-43
2.750											11-44	12-44
2.812											11-45	12-45
2.875											11-46	12-46
2.938											11-47	12-47
3.000											11-48	12-48

Dash numbers above heavy line are for short length studs per figure 2
Dash numbers below heavy line are for long length studs per figure 1

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REQUIREMENTS:

1. Materials: Steel, alloy, composition 4130 (UNS G41300) per FED-STD-66. Steel, corrosion resistant, composition 303 (UNS S30300) per FED-STD-66.

2. Protective coating: Alloy steel shall be cadmium plated in accordance with QQ-P-416, type II, class 3. Corrosion resistant steel shall be cleaned, descaled, and passivated in accordance with ASTM A380.

3. Hardness: Alloy steel shall be 26-32HRC. Corrosion resistant steel shall be 171HB minimum.

4. Threads Threads shall be in accordance with FED-STD-H28/2.

5. Push-out and torque-out values: Push-out and torque-out values shall be as specified in table IV when installed in specified material.

6. Part number: The part number consists of the letter "M", the basic specification number, thread size and length dash number (table II) and either letter "C" for corrosion resistant steel or blank for alloy steel.

Example: M63540/2- 09-20 C

- M63540/2-: Basic number of specification sheet
- 09-20: Thread size and length dash number ("E" in place of dash for short lengths)
- C: Material code (Blank if alloy steel)

INSTALLATION REQUIREMENTS:

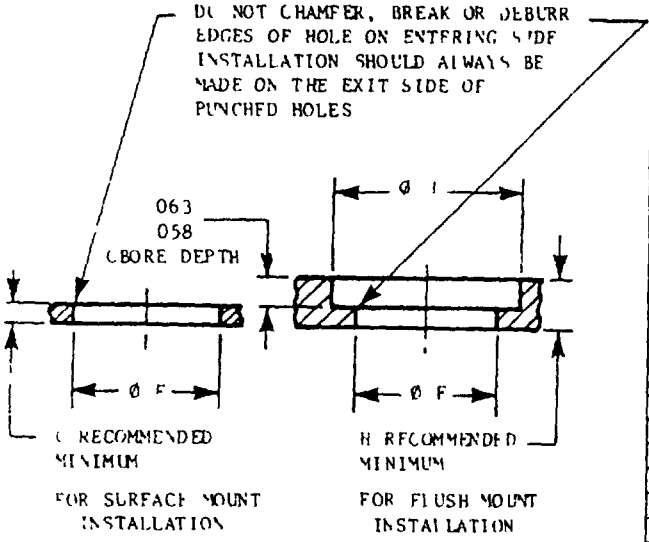


FIGURE 3 Hole preparation

TABLE III Hole dimensions and material thickness

THREAD SIZE DASH NUMBER	Ø F 1/ +.005 -.000	C 2/ MIN SURFACE MOUNT	FLUSH MOUNT H MIN	FLUSH MOUNT Ø J ±.005
01	.086	.035	.098	.187
02				
03	.113	.035	.098	.187
04				
05	.140	.035	.098	.218
06				
07	.166	.045	.118	.242
08				
09	.191	.045	.118	.281
10				
11	.250	.060	.123	.343
12				

1/ Drill or punch hole "F" diameter. For flush installation, counterbore "J" diameter to .058-.063 depth.

2/ Any sheet thickness above the minimum indicated for a specific stud diameter may be used.

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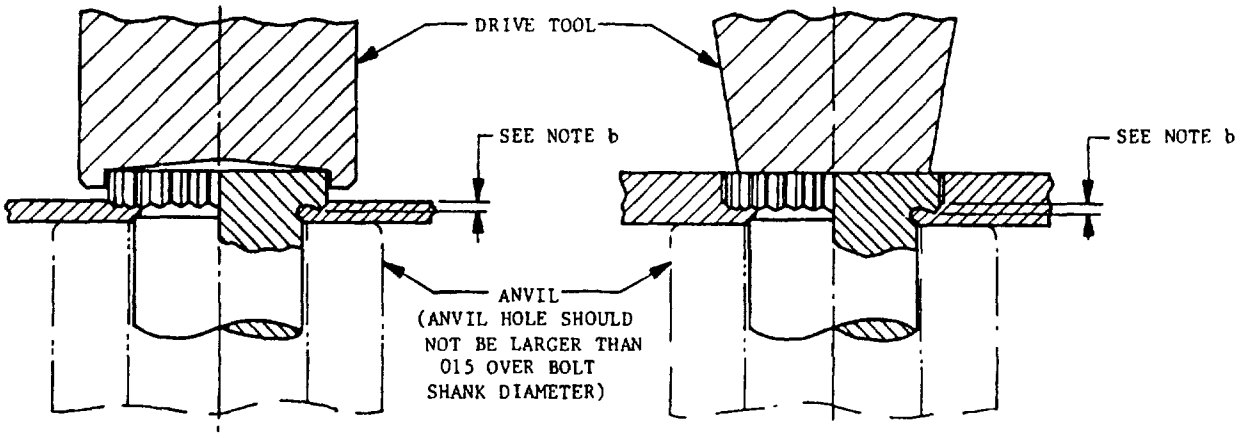


FIGURE 4. Surface mount installation

FIGURE 5. Flush mount installation

INSTALLATION PROCEDURES:

a. Insert shank of stud into "F" diameter hole.
b. Using drive tool and suitable anvil, apply force sufficient to cause a .015-.028 penetration of stud into parent material.
c. Use of manufacturers installation tools mandatory (Rosan, Inc. Newport Beach, CA - FSCM 83324).

TABLE IV: Push-out and torque-out values

THREAD SIZE DASH NUMBER	5052-H34 ALUMINUM SHEET	
	PUSH-OUT LBS MIN	TORQUE-OUT IN/LBS MIN
01 02	100	4
03 04	106	9
05 06	122	20
07 08	216	33
09 10	233	45
11 12	310	80

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AV, AT
Air Force - 82, 11
NSA - NS
DLA - IS

User Activities:
Army - MI, ME
Navy - SH

Preparing activity:
Army - AR

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